

Lecture 12

Tensors and the Tensor Product

Lecture 11 introduced the indices; this lecture builds the objects they live on. The *tensor product* $V \otimes W$ is the vector space whose elements combine a vector from V with one from W multiplicatively, and tensors of higher type unify vectors, covectors, operators, and bilinear forms into one framework. Its matrix incarnation is the *Kronecker product*, and its physical meaning is decisive: the state space of a composite quantum system is the tensor product of its parts, and *entanglement* is precisely a state that fails to factor. This closes Part II of the course.

Learning objectives

By the end of this lecture you will be able to:

- ▶ Construct $V \otimes W$ via its basis and universal property; spot non-factoring tensors.
- ▶ Compute Kronecker products and use $(A \otimes B)(v \otimes w) = (Av) \otimes (Bw)$.
- ▶ Read a type- (p, q) tensor as a multilinear map and contract indices.
- ▶ Distinguish product from entangled states in composite quantum systems.

1 The tensor product

A bilinear map $V \times W \rightarrow U$ is awkward—it is not linear. The tensor product repackages it as an ordinary linear map out of a single space.

Definition 12.1 (Tensor product). The *tensor product* $V \otimes W$ is the vector space with basis $\{e_i \otimes f_j\}$, where e_i is a basis of V and f_j a basis of W . Thus

$$\dim(V \otimes W) = (\dim V)(\dim W).$$

For $v \in V, w \in W$ the *elementary* (or decomposable) tensor $v \otimes w$ is bilinear: $(v + v') \otimes w = v \otimes w + v' \otimes w$ and $(av) \otimes w = a(v \otimes w) = v \otimes (aw)$.

A general element of $V \otimes W$ is a sum $\sum_{ij} \tau^{ij} e_i \otimes f_j$ of elementary tensors; crucially, most tensors are *not* themselves elementary—this is the algebraic seed of entanglement.

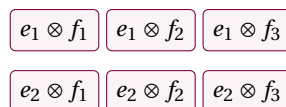


Figure 1: A basis of $V \otimes W$ for $\dim V = 2, \dim W = 3$: every pair (e_i, f_j) contributes one basis tensor, giving $\dim(V \otimes W) = 2 \cdot 3 = 6$.

Example 12.2 (A tensor that does not factor). In $\mathbb{C}^2 \otimes \mathbb{C}^2$ the tensor $e_1 \otimes e_1 + e_2 \otimes e_2$ is not elementary: if it equalled $v \otimes w$ with $v = (a, b), w = (c, d)$, its components would be the rank-one array $\begin{pmatrix} ac & ad \\ bc & bd \end{pmatrix}$, whereas the given tensor has components $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, of rank two. A tensor is decomposable exactly when its component matrix has rank one. ◀

Example 12.3 (Functions of two variables). The tensor product of polynomial spaces realizes products of functions: $\mathcal{P}_m(x) \otimes \mathcal{P}_n(y)$ is spanned by $x^i \otimes y^j$, identified with the monomials

$x^i y^j$. A general element—a polynomial in x and y —is a sum of products $f(x)g(y)$, and is a single product exactly when it factors. $\triangleleft$

Remark 12.4 (The universal property). The defining feature of $V \otimes W$ is that every bilinear map $B: V \times W \rightarrow U$ factors uniquely as $B(v, w) = \tilde{B}(v \otimes w)$ for a unique linear $\tilde{B}: V \otimes W \rightarrow U$. The tensor product is the universal recipient of bilinear maps—the precise sense in which it “linearizes multiplication.”

Remark 12.5 (Many factors). The tensor product is associative, $U \otimes (V \otimes W) = (U \otimes V) \otimes W$, so multi-factor products $V_1 \otimes \cdots \otimes V_p$ need no parentheses; their basis is the products $e_{i_1} \otimes \cdots \otimes e_{i_p}$ and their dimension is the product of the dimensions. This is the home of (p, q) tensors and of many-body quantum states.

2 The Kronecker product

In coordinates the tensor product is the *Kronecker product*: stacking the basis $e_i \otimes f_j$ lexicographically, the column of $v \otimes w$ is the Kronecker product of the columns of v and w , and an operator $A \otimes B$ has the block form

$$A \otimes B = \begin{pmatrix} A_{11}B & \cdots & A_{1n}B \\ \vdots & & \vdots \\ A_{n1}B & \cdots & A_{nn}B \end{pmatrix},$$

each entry A_{ij} of A replaced by the block $A_{ij}B$.

Proposition 12.6 (Properties of $A \otimes B$). The Kronecker product satisfies

$$(A \otimes B)(v \otimes w) = (Av) \otimes (Bw), \quad (A \otimes B)(C \otimes D) = (AC) \otimes (BD),$$

together with $(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger$, $\text{tr}(A \otimes B) = (\text{tr } A)(\text{tr } B)$, and if $Av = \lambda v$, $Bw = \mu w$ then $(A \otimes B)(v \otimes w) = \lambda\mu(v \otimes w)$.

Proof. By linearity it suffices to check the identities on elementary tensors. From the block form, $(A \otimes B)(e_i \otimes f_j) = \sum_{a,b} A_{ai} B_{bj} e_a \otimes f_b = (Ae_i) \otimes (Bf_j)$, so $(A \otimes B)(v \otimes w) = (Av) \otimes (Bw)$. The composition and eigenvalue identities follow at once: $(A \otimes B)(C \otimes D)(v \otimes w) = (ACv) \otimes (BDw) = ((AC) \otimes (BD))(v \otimes w)$, and $(A \otimes B)(v \otimes w) = (\lambda v) \otimes (\mu w) = \lambda\mu(v \otimes w)$ when $Av = \lambda v$, $Bw = \mu w$. For the adjoint, using $\langle v \otimes w, v' \otimes w' \rangle = \langle v, v' \rangle \langle w, w' \rangle$ and $\langle Av, x \rangle = \langle v, A^\dagger x \rangle$,

$$\begin{aligned} \langle (A \otimes B)(v \otimes w), x \otimes y \rangle &= \langle Av, x \rangle \langle Bw, y \rangle = \langle v, A^\dagger x \rangle \langle w, B^\dagger y \rangle \\ &= \langle v \otimes w, (A^\dagger \otimes B^\dagger)(x \otimes y) \rangle, \end{aligned}$$

so $(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger$. Finally, summing the diagonal of the block form, $\text{tr}(A \otimes B) = \sum_{a,b} A_{aa} B_{bb} = (\text{tr } A)(\text{tr } B)$. $\blacksquare$

So eigenvalues multiply: the spectrum of $A \otimes B$ is $\{\lambda_i \mu_j\}$, and if A, B are each diagonalizable (or normal) then so is $A \otimes B$.

Example 12.7 (A Kronecker product of Pauli matrices). With $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$,

$$\sigma_x \otimes \sigma_x = \begin{pmatrix} 0 & \sigma_x \\ \sigma_x & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix},$$

the operator that flips both qubits. Its eigenvalues are the products $(\pm 1)(\pm 1)$, namely $+1$ (twice, on $|+\rangle \otimes |+\rangle$ and $|-\rangle \otimes |-\rangle$) and -1 (twice, on $|+\rangle \otimes |-\rangle$ and $|-\rangle \otimes |+\rangle$)—consistent with $\text{tr}(\sigma_x \otimes \sigma_x) = (\text{tr} \sigma_x)^2 = 0$. $\triangleleft$

Example 12.8 (Kronecker product of vectors). For $v = (1, 2)$ and $w = (3, 4)$,

$$v \otimes w = (1 \cdot w, 2 \cdot w) = (3, 4, 6, 8),$$

the four products $v_i w_j$ in lexicographic order. The two-qubit state $|0\rangle \otimes |1\rangle$ is likewise the column $(0, 1, 0, 0) = |01\rangle$. $\triangleleft$

Example 12.9 (Acting factorwise). With $A = \sigma_x$, $B = \sigma_z$ and $|00\rangle = |0\rangle \otimes |0\rangle$, $(\sigma_x \otimes \sigma_z)|00\rangle = (\sigma_x|0\rangle) \otimes (\sigma_z|0\rangle) = |1\rangle \otimes |0\rangle = |10\rangle$: each factor acts on its own slot, the content of $(A \otimes B)(v \otimes w) = (Av) \otimes (Bw)$. $\triangleleft$

Example 12.10 (A single-qubit gate on two qubits). To apply a Hadamard H to the first qubit only, use $H \otimes I$: it acts as H on the first slot and leaves the second alone, $(H \otimes I)|0x\rangle = |+\rangle \otimes |x\rangle$. Local gates are tensor products with identities, and they cannot create entanglement. $\triangleleft$

Example 12.11 (A two-qubit gate matrix). The Hadamard on both qubits is

$$H \otimes H = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix},$$

each scalar entry $\frac{1}{\sqrt{2}} H_{ij}$ scaling the block $\frac{1}{\sqrt{2}} H$. Applied to $|00\rangle$ it produces the uniform superposition $\frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$. $\triangleleft$

$A_{11}B$	$A_{12}B$
$A_{21}B$	$A_{22}B$

Figure 2: The Kronecker product $A \otimes B$ for a 2×2 matrix A : each scalar entry A_{ij} becomes the block $A_{ij}B$, building an $(nm) \times (nm)$ matrix from $n \times n$ and $m \times m$ pieces.

Example 12.12 (The mixed-product rule). The rule $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$ lets gates compose factorwise. For instance $(\sigma_x \otimes I)(\sigma_z \otimes I) = (\sigma_x \sigma_z) \otimes I = (-i\sigma_y) \otimes I$, so applying σ_z then σ_x to the first qubit is the single gate $-i\sigma_y \otimes I$. $\triangleleft$

Remark 12.13 (The swap operator). The swap $|ij\rangle \mapsto |ji\rangle$ exchanges the two factors of $\mathcal{H} \otimes \mathcal{H}$. It acts as $+1$ on the symmetric tensors $S^2\mathcal{H}$ and -1 on the antisymmetric tensors $\Lambda^2\mathcal{H}$ (both defined in §4 below), so the $S^2 \oplus \Lambda^2$ split is exactly its eigenspace decomposition.

Example 12.14 (Single-qubit operators on two qubits). On two qubits $\sigma_z \otimes I$ and $I \otimes \sigma_z$ are the diagonal matrices

$$\sigma_z \otimes I = \text{diag}(1, 1, -1, -1), \quad I \otimes \sigma_z = \text{diag}(1, -1, 1, -1),$$

measuring the first and second spin respectively. They commute, and their product is $\sigma_z \otimes \sigma_z = \text{diag}(1, -1, -1, 1)$, the parity. $\triangleleft$

Remark 12.15 (Trace, determinant, and size). From $\text{tr}(A \otimes B) = \text{tr} A \text{tr} B$ and the multiplied eigenvalues, $\det(A \otimes B) = (\det A)^m (\det B)^n$ for A of size n and B of size m . The dimensions check out: $A \otimes B$ is $nm \times nm$, matching $\dim(V \otimes W) = nm$.

Remark 12.16 (Inner product on a tensor product). If V, W are inner product spaces, $V \otimes W$ inherits the inner product $\langle v \otimes w, v' \otimes w' \rangle = \langle v, v' \rangle \langle w, w' \rangle$ and an orthonormal basis $\{e_i \otimes f_j\}$. Composite quantum systems are therefore again inner product (Hilbert) spaces, so everything in Lectures 7–10 applies to them.

3 Tensors as multilinear maps

A *tensor of type (p, q)* is a multilinear map taking p covectors and q vectors to a scalar; equivalently it is an element of $V^{\otimes p} \otimes (V^*)^{\otimes q}$, with components $\tau^{i_1 \dots i_p}_{j_1 \dots j_q}$ carrying p upper and q lower indices. The familiar objects are low-rank tensors: a vector is $(1, 0)$, a covector $(0, 1)$, a bilinear form (such as the metric g_{ij}) is $(0, 2)$, and a linear operator is $(1, 1)$.

Proposition 12.17 (Operators are tensors). There is a canonical isomorphism

$$\mathcal{L}(V, W) \cong W \otimes V^*, \quad A^a{}_i \mapsto A^a{}_i w_a \otimes e^i,$$

sending an operator to the tensor built from its matrix. The identity operator corresponds to $\delta^a{}_i w_a \otimes e^i$ —the Kronecker delta is the identity tensor.

Proof. Define $\Psi: W \otimes V^* \rightarrow \mathcal{L}(V, W)$ on elementary tensors by $\Psi(w \otimes \varphi)(v) = \varphi(v) w$, extended linearly; this uses only the pairing of V^* with V , so it needs no basis. On the basis $\{w_a \otimes e^i\}$, $\Psi(w_a \otimes e^i)(e_j) = e^i(e_j) w_a = \delta^i_j w_a$, the elementary operator $E^a{}_i$ sending $e_i \mapsto w_a$ and $e_j \mapsto 0$ ($j \neq i$). As $\{E^a{}_i\}$ is a basis of $\mathcal{L}(V, W)$, Ψ carries a basis to a basis and is an isomorphism, with $\dim = (\dim W)(\dim V)$. Its inverse sends $A = A^a{}_i$ to $A^a{}_i w_a \otimes e^i$, the correspondence of the statement; that tensor returns $v \mapsto A^a{}_i e^i(v) w_a = A^a{}_i v^i w_a = Av$. ■

Example 12.18 (Outer products are decomposable operators). The rank-one operator $|a\rangle\langle b|$ corresponds to the elementary tensor $|a\rangle \otimes \langle b| \in V \otimes V^*$. A general operator is a sum of such outer products (its tensor is a sum of decomposables), and it is itself decomposable—rank one—exactly when it is a single outer product $|a\rangle\langle b|$. ◀

Example 12.19 (The metric as a $(0, 2)$ tensor). The inner product is the $(0, 2)$ tensor $g = g_{ij} e^i \otimes e^j$ with $g_{ij} = \langle e_i, e_j \rangle$; it eats two vectors and returns $g(u, v) = g_{ij} u^i v^j$. Lowering an index, $v_j = g_{ij} v^i$, is the contraction of g with a vector—the index-lowering of Lecture 11. ◀

Example 12.20 (An operator as components). The operator $T = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$ has $(1, 1)$ components $T^1{}_1 = 2, T^1{}_2 = 1, T^2{}_1 = 0, T^2{}_2 = 3$, and as a tensor $T = 2 e_1 \otimes e^1 + e_1 \otimes e^2 + 3 e_2 \otimes e^2$. Its trace is the contraction $T^i{}_i = 2 + 3 = 5$. ◀

Example 12.21 (Trace via the identity tensor). Contracting the identity tensor $\delta^i{}_j$ with itself gives $\delta^i{}_i = \sum_i 1 = n = \dim V$, so $\text{tr } I = \dim V$. The Kronecker delta is the identity operator, and contracted it is the dimension. ◀

The single operation that lowers tensor rank is *contraction*: summing over a matched upper and lower index, $\tau^i{}_i$, which for a $(1, 1)$ tensor is exactly the trace. Under a change of basis each upper index transforms with P^{-1} and each lower index with P (Lecture 11), so a contracted pair is invariant.

Example 12.22 (Contraction is the trace). For an operator $T = T^i{}_j e_i \otimes e^j$, contracting the upper against the lower index gives $T^i{}_i = \sum_i T^i{}_i = \text{tr } T$, basis-independent by Lecture 11. More generally, contracting one pair of a higher tensor lowers its type by $(1, 1)$, turning, say, a $(1, 2)$ tensor into a covector. ◀

Example 12.23 (A transformation law). A $(1, 1)$ tensor transforms as $T'^i{}_j = (P^{-1})^i{}_k T^k{}_l P^l{}_j$, a $(0, 2)$ tensor as $g'_{ij} = P^k{}_i P^l{}_j g_{kl}$, and a $(2, 0)$ tensor with two factors of P^{-1} . The rule is mechanical: one factor of P^{-1} per upper index, one of P per lower index. ◀

Example 12.24 (A $(2, 0)$ tensor). An antisymmetric $(2, 0)$ tensor $\omega^{ij} = -\omega^{ji}$ is a *bivector*; in $\mathbb{R}^3$ its three independent components $\omega^{12}, \omega^{23}, \omega^{31}$ are those of the cross product, which is why the cross product is really an antisymmetric tensor wearing the costume of a vector. ◀

4 Symmetric and antisymmetric tensors

Within $V \otimes V$ live two natural subspaces.

Definition 12.25 (Symmetric and antisymmetric tensors). A tensor in $V \otimes V$ is *symmetric* if swapping its two slots leaves it unchanged and *antisymmetric* if swapping flips the sign. These form subspaces S^2V and Λ^2V , and

$$V \otimes V = S^2V \oplus \Lambda^2V.$$

They are built from vectors by the symmetrized product $v \otimes w + w \otimes v$ and the wedge $v \wedge w = v \otimes w - w \otimes v$.

The wedge product is alternating ($v \wedge v = 0$) and feeds the theory of determinants and differential forms (Lecture 11). The split mirrors the decomposition of a square matrix into symmetric and antisymmetric parts. In physics the symmetric tensors describe bosons (symmetric multi-particle states) and the antisymmetric ones describe fermions (the Pauli exclusion principle is the antisymmetry $v \wedge v = 0$).

Example 12.26 (Splitting a tensor). Any $\tau = \sum \tau^{ij} e_i \otimes e_j$ splits as $\tau = \tau_+ + \tau_-$ with symmetric part $\tau_+^{ij} = \frac{1}{2}(\tau^{ij} + \tau^{ji})$ and antisymmetric part $\tau_-^{ij} = \frac{1}{2}(\tau^{ij} - \tau^{ji})$. For $e_1 \otimes e_2$ this gives $\frac{1}{2}(e_1 \otimes e_2 + e_2 \otimes e_1) + \frac{1}{2}e_1 \wedge e_2$ —a symmetric piece plus a wedge. ◀

Example 12.27 (Symmetric tensors are polynomials). Symmetric tensors in $S^k V^*$ correspond to homogeneous degree- k polynomials in the coordinates, $e^i \otimes_S e^j \leftrightarrow x_i x_j$. Their dimension $\binom{n+k-1}{k}$ counts degree- k monomials in n variables: symmetric tensor algebra is polynomial algebra. ◀

Remark 12.28 (Identical particles). Quantum statistics is the symmetry of multi-particle tensors: n identical bosons live in the symmetric power $S^n \mathcal{H}$, n fermions in $\Lambda^n \mathcal{H}$. Since Λ^n vanishes once n exceeds $\dim \mathcal{H}$, no two fermions share a state—the Pauli principle, restated as a fact about antisymmetric tensors.

Remark 12.29 (Dimensions). For $\dim V = n$, $\dim S^2V = \binom{n+1}{2}$ and $\dim \Lambda^2V = \binom{n}{2}$, summing to $n^2 = \dim(V \otimes V)$. At the top, $\Lambda^n V$ is one-dimensional—its single basis element is the determinant.

Example 12.30 (The wedge and the determinant). In $\Lambda^2 \mathbb{R}^2$, $u \wedge v = (u_1 v_2 - u_2 v_1) e_1 \wedge e_2$: the lone coefficient is the 2×2 determinant. In general

$$v_1 \wedge \cdots \wedge v_n = \det[v_1 \cdots v_n] (e_1 \wedge \cdots \wedge e_n),$$

so the determinant is the component of the top wedge—volume as an antisymmetric tensor. ◀

Example 12.31 (Two-forms in three dimensions). $\Lambda^2 \mathbb{R}^3$ has dimension $\binom{3}{2} = 3$, with basis $e_1 \wedge e_2, e_2 \wedge e_3, e_3 \wedge e_1$ —the same count as vectors in $\mathbb{R}^3$. This coincidence is why the cross

product exists only in three dimensions: there a bivector $u \wedge v$ can be reinterpreted as a vector. $\triangleleft$

5 Tensor products in quantum mechanics

The state space of a composite system is the tensor product of the parts: $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$. For two qubits, $\mathcal{H} = \mathbb{C}^2 \otimes \mathbb{C}^2 = \mathbb{C}^4$ with orthonormal basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ (writing $|ij\rangle = |i\rangle \otimes |j\rangle$).

Definition 12.32 (Product and entangled states). A state of $\mathcal{H}_A \otimes \mathcal{H}_B$ is a *product state* if it factors as $|\psi\rangle \otimes |\phi\rangle$, and *entangled* otherwise.

The Bell state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ is entangled: no choice of single-qubit states $|\psi\rangle, |\phi\rangle$ gives $|\psi\rangle \otimes |\phi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. Operators on the composite are built as $A \otimes B$ (act with A on the first factor, B on the second), though—crucially—not every operator is of this form, which is exactly how entangling gates generate entanglement from product inputs.

Example 12.33 (The Bell state is entangled). Suppose $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = (a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle) = ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle$. Matching coefficients forces $ad = bc = 0$ yet $ac = bd = \frac{1}{\sqrt{2}} \neq 0$, a contradiction. No product factorization exists—the state is entangled. $\triangleleft$

Example 12.34 (A state that does factor). By contrast $\frac{1}{\sqrt{2}}(|00\rangle + |01\rangle) = |0\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |0\rangle \otimes |+\rangle$: its coefficient grid $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ has rank one, so it is a product state, not entangled. $\triangleleft$

Example 12.35 (A decomposable tensor's grid). The product $(|0\rangle + |1\rangle) \otimes (|0\rangle + 2|1\rangle)$ expands to $|00\rangle + 2|01\rangle + |10\rangle + 2|11\rangle$, with coefficient grid

$$\begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \end{pmatrix},$$

manifestly rank one—the algebraic signature of a product state. $\triangleleft$

Example 12.36 (Creating entanglement with a gate). A Hadamard on the first qubit of $|00\rangle$ gives the product state $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |0\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$; the controlled-NOT $|1x\rangle \mapsto |1\bar{x}\rangle$ then yields $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. A product input has become the entangled Bell state. $\triangleleft$

Example 12.37 (The CNOT matrix). In the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ the controlled-NOT is

$$U_{\text{CN}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix},$$

the identity on the first two basis states and a swap on the last two. It is *not* of the form $A \otimes B$, which is precisely why it can entangle. $\triangleleft$

Example 12.38 (Singlet and triplet). Two spin- $\frac{1}{2}$ particles combine in $\mathbb{C}^2 \otimes \mathbb{C}^2$ into the anti-symmetric *singlet* $\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$ and three symmetric *triplet* states. The singlet is maximally entangled, and the symmetry/antisymmetry under particle exchange is the $S^2 \oplus \Lambda^2$ split of Definition 12.25 made physical. $\triangleleft$

Example 12.39 (Three-qubit entanglement). The GHZ state $\frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$ lives in $(\mathbb{C}^2)^{\otimes 3} = \mathbb{C}^8$ and is genuinely three-party entangled: no splitting into a single qubit and a pair reproduces it. Composite systems pile up one tensor factor per subsystem. $\triangleleft$

Example 12.40 (A correlation). On the Bell state, the joint observable $\sigma_z \otimes \sigma_z$ has expectation +1—measuring σ_z on both qubits always gives equal outcomes—while the local expectations $\sigma_z \otimes I$ and $I \otimes \sigma_z$ are each 0. Perfect correlation with no local information: entanglement in a number. ◀

Example 12.41 (Correlation in the x -basis). On $|\Phi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, the flip-both operator $\sigma_x \otimes \sigma_x$ of Example 12.7 gives

$$(\sigma_x \otimes \sigma_x)|\Phi\rangle = \frac{1}{\sqrt{2}}(|11\rangle + |00\rangle) = |\Phi\rangle,$$

so $\langle\Phi|\sigma_x \otimes \sigma_x|\Phi\rangle = 1$: the Bell state is perfectly correlated in the x -basis as well as the z -basis. ◀

Remark 12.42 (The Bell basis). The four Bell states $\frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle)$ and $\frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle)$ form an orthonormal basis of $\mathbb{C}^2 \otimes \mathbb{C}^2$ made entirely of maximally entangled states. A measurement in this basis—a *Bell measurement*—is the key step of teleportation and superdense coding.

Example 12.43 (The four Bell states). The Bell basis of $\mathbb{C}^2 \otimes \mathbb{C}^2$ is

$$\begin{aligned} |\Phi^\pm\rangle &= \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle), \\ |\Psi^\pm\rangle &= \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle), \end{aligned}$$

four orthonormal, maximally entangled states. Every two-qubit state expands in this basis, and a Bell measurement projects onto one of them. ◀

Remark 12.44 (Teleportation and no cloning). A shared Bell pair lets one party send an unknown qubit using only a Bell measurement and two classical bits—entanglement plus the tensor-product structure moves quantum information without moving the particle (and not faster than light, since the classical bits are needed). Conversely, no linear U can satisfy $U(|\psi\rangle \otimes |0\rangle) = |\psi\rangle \otimes |\psi\rangle$ for all $|\psi\rangle$: unknown states cannot be cloned, a structural fact about operators on $\mathcal{H} \otimes \mathcal{H}$.

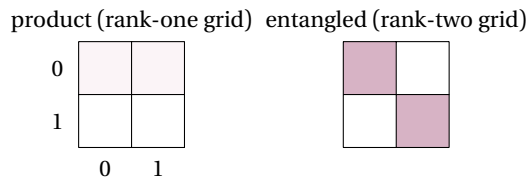


Figure 3: Coefficient grids τ^{ij} of a two-qubit state. A product state's grid is rank one (an outer product of single-qubit amplitudes); the Bell state's grid is rank two and cannot be factored—the visual signature of entanglement.

Remark 12.45 (Reduced states and the partial trace). A subsystem of an entangled state has no pure state of its own; it is described by a *reduced density matrix*, the *partial trace* over the other factor (a contraction on one tensor slot). For a Bell state each qubit's reduced state is the maximally mixed $\frac{1}{2}I$ —maximal ignorance of the part despite full knowledge of the whole, the hallmark of entanglement.

Example 12.46 (A reduced density matrix). For the Bell state $|\Phi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, tracing out the second qubit gives $\rho_A = \text{tr}_B|\Phi\rangle\langle\Phi| = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{1}{2}I$ on the first qubit. The part is maximally mixed even though the whole is a definite pure state—the defining tension of entanglement. ◀

Remark 12.47 (Schmidt decomposition). Every pure state of $\mathcal{H}_A \otimes \mathcal{H}_B$ can be written $\sum_k \lambda_k |a_k\rangle \otimes |b_k\rangle$ with orthonormal $|a_k\rangle, |b_k\rangle$ and $\lambda_k \geq 0$ (the *Schmidt decomposition*, the tensor-product cousin of the SVD). The count of nonzero λ_k is the Schmidt rank: 1 for product states, larger for entangled ones.

Remark 12.48 (Exponential growth). An n -qubit system has state space $(\mathbb{C}^2)^{\otimes n} = \mathbb{C}^{2^n}$: the dimension doubles with each qubit. This exponential growth is both the promise of quantum computing—a state holds 2^n amplitudes—and the obstacle to simulating quantum systems on classical hardware.

Summary of main concepts

The tensor product $V \otimes W$ has basis $\{e_i \otimes f_j\}$ and dimension $(\dim V)(\dim W)$; its elements are sums of elementary tensors $v \otimes w$, most of which do not factor. In coordinates this is the Kronecker product, with $(A \otimes B)(v \otimes w) = (Av) \otimes (Bw)$, $\text{tr}(A \otimes B) = \text{tr } A \text{ tr } B$, and eigenvalues $\lambda\mu$. A type- (p, q) tensor is a multilinear map with p upper and q lower indices, unifying vectors, covectors, the metric, and operators ($\mathcal{L}(V, W) \cong W \otimes V^*$); contraction over a matched pair generalizes the trace. Within $V \otimes V$, the symmetric and antisymmetric tensors split the space and model bosons and fermions. In quantum mechanics composite systems live in $\mathcal{H}_A \otimes \mathcal{H}_B$, and entanglement is the existence of states that do not factor.

- ▶ **Tensor product** $V \otimes W$ —basis $\{e_i \otimes f_j\}$, dimension $(\dim V)(\dim W)$; most elements do not factor.
- ▶ **Kronecker product**—the coordinate form; $\text{tr}(A \otimes B) = \text{tr } A \text{ tr } B$, eigenvalues $\lambda\mu$, size $nm \times nm$.
- ▶ **Type- (p, q) tensor**—a multilinear map with p upper and q lower indices; unifies vectors, covectors, the metric, and operators ($\mathcal{L}(V, W) \cong W \otimes V^*$); contraction generalizes the trace.
- ▶ **Symmetric and antisymmetric tensors**—split $V \otimes V$; model bosons and fermions.
- ▶ **Composite systems and entanglement**— $\mathcal{H}_A \otimes \mathcal{H}_B$; entangled states are those that do not factor.

Exercises for this lecture are in the companion sheet `exercises-12.pdf`.